

Layered singular-cardinal groups are ambitable

Candidate partial result for arXiv:2204.09427

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Status. Candidate substantial partial result, likely valid; human review is needed. The unrestricted precompact-or-ambitable and Samuel-separation questions remain open.

1 Source questions and current status

Schneider records the open question whether every topological group is either precompact or ambitable [1, Section 5]. In the formulation used below, a group is ambitable if, for every continuous invariant pseudometric Δ , some $h \in U_b(G)$ has

$$\text{BLip}_b(\Delta) \subseteq \overline{\text{orb}(h)}^G, \quad (1.1)$$

where the bar denotes pointwise closure and $\text{BLip}_b(\Delta)$ is the ball of real-valued functions f with $\|f\|_\infty \leq 1$ and $|f(x) - f(y)| \leq \Delta(x, y)$. This is the standard Lipschitz-ball characterization used by Pachl [3, 4].

$$\stackrel{(4)}{=} \frac{1}{2} \left(\mu \left(\left(0 - \frac{1}{2} \right)^2 \right) + \mu \left(\left(1 - \frac{1}{2} \right)^2 \right) \right) = \frac{1}{4}.$$

This proves (5) and thus completes the argument. $\square$

Contemplating the hypothesis of Theorem 5.1, we now turn to a certain class of topological groups. Following Pachl [37], a topological group G will be called *ambitable*⁶ if

$$\forall B \in \text{RUEB}(G) \exists f \in \text{RUCB}(G): \quad B \subseteq \{\Phi_\xi f \mid \xi \in S(G)\}.$$

Precompact topological groups are non-ambitable [37, Theorem 2]. It is an open problem [37, Question 1] whether the converse is true as well, i.e., whether every topological group is either precompact or ambitable. Within the class of locally compact groups, this question has been answered in the affirmative [37, Corollary 15]. Another related result is the following.

Remark 5.2. Every separable topological group is either precompact or ambitable. This is a direct consequence of [37, Corollary 16].

For our purposes, the following basic observation about ambitable topological groups will be relevant.

Lemma 5.3. *If G is an ambitable topological group, then $S(G)$ admits a continuum of pairwise disjoint, closed, G -invariant, non-empty subsets.*

Figure 1: The precompact-or-ambitable problem in the source paper [1, p. 27].

The same section asks whether the Samuel compactification of every nonprecompact group contains two disjoint closed invariant nonempty subsets.

$$(g\xi)(f) = \xi(f \circ \lambda_g) = (\Phi_\xi f)(g) = i$$

for all $g \in G$, thus $\xi \in T_i$ and so $T_i \neq \emptyset$. This completes the argument. $\square$

Quantitative improvements of Lemma 5.3 are known for non-compact, locally compact groups [25, Theorem 4.1], as well as for non-precompact, separable topological groups [12, Theorem 1.3] (see also [21, Theorem 3.8]). It is an open problem whether the Samuel compactification of every non-precompact topological contains two disjoint closed, invariant, non-empty subsets [21, paragraph after Theorem 3.8].

Regarding the notation used in the following two corollaries, we refer to

Figure 2: The weaker Samuel-separation problem in the source paper [1, p. 28].

Pachl's June 2026 note explicitly says that the first question remains open and proves ambitability for one exceptional group: the finite-support $\mathbb{Z}/2$ -sum on ω_ω , layered by the cardinals $\aleph_k$ [4]. The theorem below extracts and strengthens the exact combinatorial mechanism of that construction.

Another amitable group

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Abstract

It is an open question whether every topological group is precompact or amitable. This note presents a topological group that is amitable but does not satisfy previously known sufficient conditions for being so.

1 Group G

The following question was asked in [2].

Question. *Is every topological group either precompact or amitable?*

Theorem 3.35 in [3] gave a positive answer for a large class of topological groups. That has left the question open for the remaining rather special groups. The present note describes one such group G to

Figure 3: Current-status statement and question in Pachl's June 2026 note [4, p. 1].

2 Idea of the proof

The topology has a countable sequence of projection kernels. Any continuous invariant pseudometric is dominated by a step ultrametric on these kernels, so it is enough to code finite Lipschitz patterns for one such metric. At layer $k+1$, a pattern must use a prescribed fiber in order to remain coherent with its restriction to layer k . The fiber has cardinality κ_{k+1} , while all previously placed blocks

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forbid at most κ_k points plus fewer than κ_{k+1} same-level points. Strict cardinal growth leaves a fresh translate. Coherence then makes the disjoint-block code uniformly continuous: nearby code points have the same lower projection, hence lie over the same restricted pattern. The orbit of its extension realizes all finite patterns.

This proves a broad class of new examples, but it does not solve the full source question: a general nonprecompact group need not possess coherent homomorphic projections with successively larger fibers.

3 The generalized layered theorem

Let

$$\kappa_0 < \kappa_1 < \kappa_2 < \cdots \quad (3.1)$$

be arbitrary infinite cardinals. Choose nested sets $I_0 \subset I_1 \subset \cdots$ with $|I_k| = \kappa_k$, put $I = \bigcup_k I_k$, and fix a nontrivial finite abelian group A . Write

$$G = A^{(I)}, \quad G_k = A^{(I_k)},$$

for the finite-support direct sums, and let $\pi_k : G \rightarrow G_k$ be coordinate restriction. For $x \neq 0$ define

$$m(x) := \min\{k : \text{supp}(x) \cap I_k \neq \emptyset\}. \quad (3.2)$$

Given any strictly decreasing positive sequence $s_k \rightarrow 0$, set

$$\rho(x, y) = s_{m(x-y)} \quad (x \neq y), \quad \rho(x, x) = 0. \quad (3.3)$$

Theorem 1. *The metric group (G, ρ) is nonprecompact and ambitable.*

Proof. The first occupied layer cannot move downward under addition, so ρ is an invariant ultrametric. Moreover,

$$\rho(x, y) \leq s_{k+1} \iff \pi_k(x) = \pi_k(y). \quad (3.4)$$

Step 1: reduction to a step ultrametric. Let Δ be a continuous invariant pseudometric. Replacing it by $\min\{2, \Delta\}$ does not change the required conclusion. Put $r_0 = 2$ and for $k \geq 1$ put

$$r_k := \sup\{\Delta(0, x) : \pi_{k-1}(x) = 0\}.$$

The projection kernels form a neighborhood basis at zero, so $r_k \downarrow 0$. Then $t_k = r_k + 2^{-k}$ is strictly decreasing to zero. The invariant step ultrametric

$$d(x, y) = t_{m(x-y)} \quad (x \neq y) \quad (3.5)$$

dominates Δ . Hence $\text{BLip}_b(\Delta) \subseteq \text{BLip}_b(d)$, and it suffices to construct one orbit whose pointwise closure contains $\text{BLip}_b(d)$.

Step 2: finite patterns. Fix a countable dense set $Q \subset [-1, 1]$. Let Φ be the family of nonempty finite partial functions $\varphi : G \dashrightarrow Q$ such that

1. φ is the restriction of a function with sup norm at most two and Lipschitz constant at most two for d ;
2. $\pi_j(\text{dom } \varphi) \subseteq \text{dom } \varphi$ for every j .

Put

$$\Phi_k = \{\varphi \in \Phi : \text{dom } \varphi \subseteq G_k\}.$$

Projection closure preserves finiteness: the projections of a finite-support point change only at the finitely many layers occupied by its support. Thus every finite test set can be enlarged to a finite projection-closed one. Rational approximation, followed by the real McShane extension, shows that a family realizing every $\varphi \in \Phi$ is pointwise dense in $\text{BLip}_b(d)$. Also

$$|\Phi_k| = |G_k| = \kappa_k. \quad (3.6)$$

The upper bound is finite-pattern counting. For the lower bound, associate to each $x \in G_k$ a constant-valued pattern on the finite projection closure of $\{x\}$.

Step 3: coherent disjoint translations. For $\varphi \in \Phi$, write φ/j for its restriction to G_j . We claim that translations z_φ can be chosen so that

$$\varphi \in \Phi_k \Rightarrow z_\varphi \in G_k, \quad \pi_j(z_\varphi) = z_{\varphi/j}, \quad (3.7)$$

and all finite blocks

$$B_\varphi := \text{dom}(\varphi) + z_\varphi \quad (3.8)$$

are pairwise disjoint.

At level zero, well-order Φ_0 in type κ_0 . Before any stage $\alpha < \kappa_0$, fewer than κ_0 finite blocks have been placed, and each block forbids only finitely many translations, so a new translation in G_0 remains.

Inductively suppose all patterns in Φ_k have been placed. Well-order $\Phi_{k+1} \setminus \Phi_k$ in type κ_{k+1} . For a current pattern χ , coherence requires

$$z_\chi \in F_\chi := G_{k+1} \cap \pi_k^{-1}(z_{\chi/k}). \quad (3.9)$$

Since $|I_{k+1} \setminus I_k| = \kappa_{k+1}$ and A is finite nontrivial, $|F_\chi| = \kappa_{k+1}$. Each earlier finite block forbids only finitely many members of F_χ : explicitly, an intersection $(u + z_\chi) = (v + z_\varphi)$ forces $z_\chi = v + z_\varphi - u$. At a same-level stage $\alpha < \kappa_{k+1}$, the total forbidden set therefore has cardinal at most

$$\max\{\kappa_k, |\alpha|\} < \kappa_{k+1}. \quad (3.10)$$

Choose a remaining point of F_χ . This proves (3.7)–(3.8) by recursion. No regularity or successor assumption on κ_{k+1} was used.

Step 4: uniform compatibility of the code. On the disjoint union $D = \bigcup_{\varphi \in \Phi} B_\varphi$, define

$$\tilde{h}(x + z_\varphi) = \varphi(x). \quad (3.11)$$

Suppose $d(x + z_\varphi, y + z_\psi) \leq t_{k+1}$. From (3.4), applied to d , and coherence,

$$\pi_k(x) + z_{\varphi/k} = \pi_k(y) + z_{\psi/k}. \quad (3.12)$$

The two sides lie in the translated blocks belonging to the restricted patterns. Disjointness forces $\varphi/k = \psi/k$, and then $\pi_k(x) = \pi_k(y)$. Projection closure and the two-Lipschitz property give

$$\begin{aligned} |\varphi(x) - \psi(y)| &\leq |\varphi(x) - \varphi(\pi_k x)| + |\psi(\pi_k y) - \psi(y)| \\ &\leq 2d(x, \pi_k x) + 2d(y, \pi_k y) \leq 4t_{k+1}. \end{aligned} \quad (3.13)$$

Thus $\tilde{h}$ is bounded and uniformly continuous. The standard uniform extension theorem for bounded real functions on metric subspaces extends it to $h \in U_b(G)$. For every φ , the translate $z_\varphi \cdot h$, defined by $(z \cdot h)(x) = h(x + z)$, realizes φ . Step 2 now yields

$$\text{BLip}_b(d) \subseteq \overline{\text{orb}(h)}^G.$$

Since Δ was arbitrary, G is ambitable.

Finally, $\ker \pi_0$ is open and $G/\ker \pi_0 \cong G_0$ is infinite. The corresponding right-uniform entourage has no finite cover, so G is not precompact. $\square$

Corollary 2. *For every singular cardinal κ of countable cofinality and every nontrivial finite abelian group A , there is a nonprecompact ambitable group of cardinality κ of the form in Theorem 1. Its Samuel compactification contains continuum many pairwise disjoint closed invariant nonempty subsets.*

Proof. Choose any strictly increasing cofinal sequence of infinite cardinals κ_k below κ . Then $|A^{(I)}| = |I| = \kappa$. Schneider’s Lemma 5.3 says that the Samuel compactification of an ambitable group contains a continuum of pairwise disjoint closed invariant nonempty subsets [1]. $\square$

Pachl’s 2026 example is recovered by taking $A = \mathbb{Z}/2$, $I_k = \omega_k$, and $\kappa_k = \aleph_k$. Theorem 1 allows every countable strictly growing cardinal tower and every finite nontrivial abelian coefficient group.

4 Upgrade attempts and remaining obstruction

Eight focused upgrades were audited. First, the displayed Pestov concentration problem in the source was discarded because the same paper answers it negatively. Second, the June 2026 status check confirmed that the ambitability problem itself remains open. Third, the cardinal recursion was abstracted successfully: strict growth, not regularity or a successor step, is the required resource. Fourth, the coefficient group was generalized from $\mathbb{Z}/2$ to every finite nontrivial abelian group by using general difference avoidance rather than characteristic two.

The remaining routes did not close the full problem. If consecutive layers have equal cardinality, the prescribed fiber may be exhausted by earlier patterns. With an uncountable cofinal tower, projection closure of a finite domain can become infinite, breaking the finite-pattern recursion. An arbitrary nonprecompact group supplies separated sets but not coherent homomorphic retractions, so equality of close code-point projections no longer reduces to one lower-level block. Finally, a direct attempt on the weaker Samuel problem gives the strong conclusion in Corollary 2, but outside the layered class invariant closures of two pieces of a uniformly discrete set can still meet the same minimal subflow.

These are structural failures of the present method, not counterexamples. The unrestricted questions therefore remain open.

5 Verification, novelty bounds, and review recommendation

The proof is self-contained after the standard Lipschitz-ball characterization of ambitability and the bounded uniform-extension theorem. The recursion was checked at arbitrary (including singular) layer cardinals: for every initial stage $\alpha < \kappa_{k+1}$ one has $|\alpha| < \kappa_{k+1}$, which is all (3.10) needs. Finite projection closure, fiber cardinality, the disjointness criterion, and the modulus (3.13) were checked separately. No computation is part of the proof.

The run’s cheap registry, solution, attempt, proof-gap, and ledger indexes were searched for arXiv:2204.09427, arXiv:2606.06000, “precompact or ambitable”, “singular ambitable”, and the core cardinal-tower terms. Pachl’s 4 June 2026 primary paper was also checked directly. The bounded search found the single $\mathbb{Z}/2, \aleph_\omega$ example but no arbitrary-tower or finite-coefficient statement. Novelty confidence is therefore moderate, not definitive.

Promote as a substantial partial result. Human review should focus on the finite-pattern density convention and the coherent-translation recursion, especially (3.9)–(3.13).

References

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